EXTRA TUTORIAL

1. Please write True [T] or False [F] for the following statements. [6m]

1	The smallest “unit” of data on a binary computer is a single byte	
2	2 Words = 8 nibble	
3	MSB is the leftmost bit.	
4	1001 is an invalid code in BCD	
5	Gray code is used to facilitate error correction in digital communications.	
6	Parity bit is append to the code at the rightmost position (LSB)	

2. Convert the following numbers to its decimal equivalent.

   a. $FE04_{16}$ [2m]

   b. 750_5 [2m]

3. Convert 9000_{10} to following number representations.

   a. hexacedimal number. [3m]

   b. binary number [1m]

4. Convert 564.45_{10} to following number representations. Give answer to 3 fractional point.

   a. binary number. [5m]

   b. hexadecimal number [1m]

      0010 0011 0100 . 0110

5. Convert 2288 to: [1m]

   a. BCD

   b. Gray Code [2m]

1	0	0	0	1	0	1	0	0	0	1	0	0	0
1	1	0	0	1	1	1	1	0	0	1	1	0	0

6. What is the ASCII binary code for the character 'S' (53_H) using **odd** parity code. [2m]

7. Give reasons why -8 is **invalid** in the following 4 bits signed number representation:

   a. Sign and magnitude [2m]

   b. 1's complement [2m]

8. Perform the following operations using 8-bit 2's complement signed number.

   a. 24 – 17 [3m]

   b. -17 + 24 [3m]